WAVE DRIVE TANK

Mechanism Analysis & Computational Simulation

From rotary motion → travelling sine wave → linear translation

Presented by:

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Inspired by: James Bruton — "Experimental Wave Drive V2"

Agenda

Roadmap of the project deck

01	**What is a Wave Drive?**	*Concept & inspiration*
02	**The Single Cam — Building Block**	*Eccentric circular cam → SHM*
03	**From One Cam to a Wave**	*Phase shifting N cams together*
04	**Mathematics of Motion**	*Position, velocity, acceleration equations*
05	**Why Does It Move?**	*Propulsion physics & contact analysis*
06	**MATLAB Simulation Suite**	*Five animated sub-simulations* Integration possibilities
07	**Speed & Performance Analysis**	*Operating range and trade-offs*
08	**Conclusions & Future Work**	*possibilities*

01

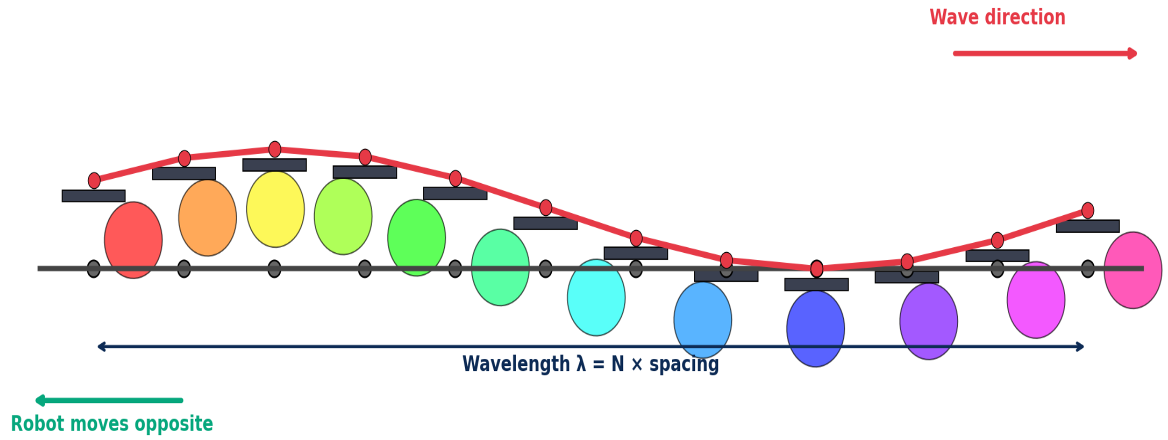
What is a Wave Drive?

Concept

The Wave Drive is an alternative locomotion mechanism that uses a **rotating camshaft** underneath the robot to generate a **travelling sine wave** of plates touching the ground.

As the wave moves backward along the chassis, the ground exerts a forward friction force → **the robot translates.**

N Eccentric Cams (each phase-shifted by 360°/N) → Travelling Sine Wave

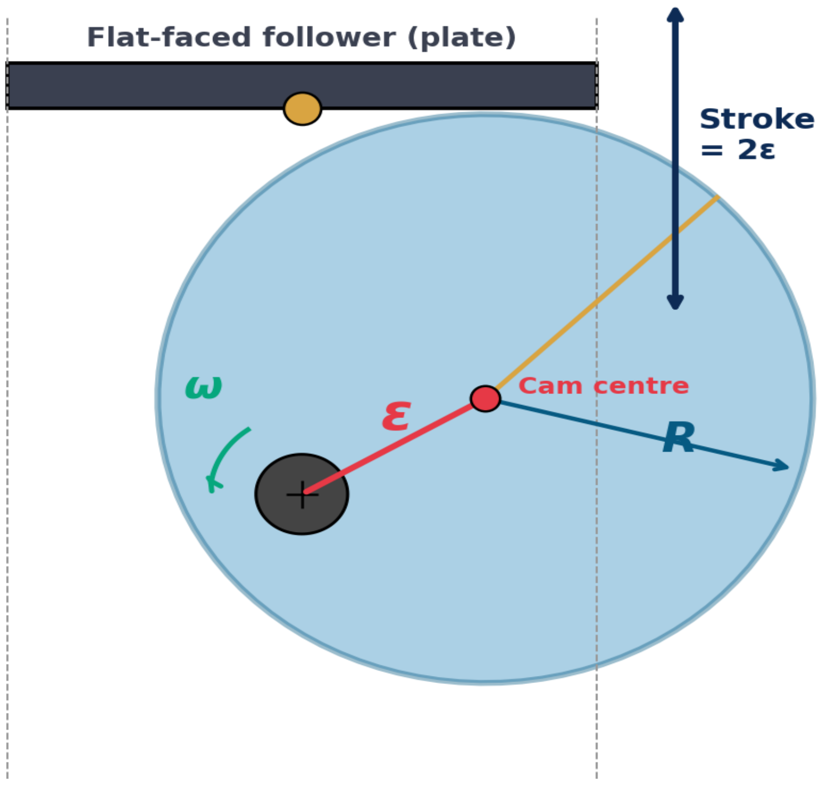


Key Insight

Each plate moves only up & down (SHM), but the COLLECTION of plates phase-shifted across the tank creates a wave that travels horizontally.

- ✓ Distributes contact load over many plates
- ✓ Excellent for soft / uneven terrain
- ✓ Speed scales linearly with shaft RPM
- ✓ Single motor input drives all plates

The Single Cam — The Building Block



Shaft axis

Eccentric Circular Cam

A perfect circle, mounted off-centre on the shaft.

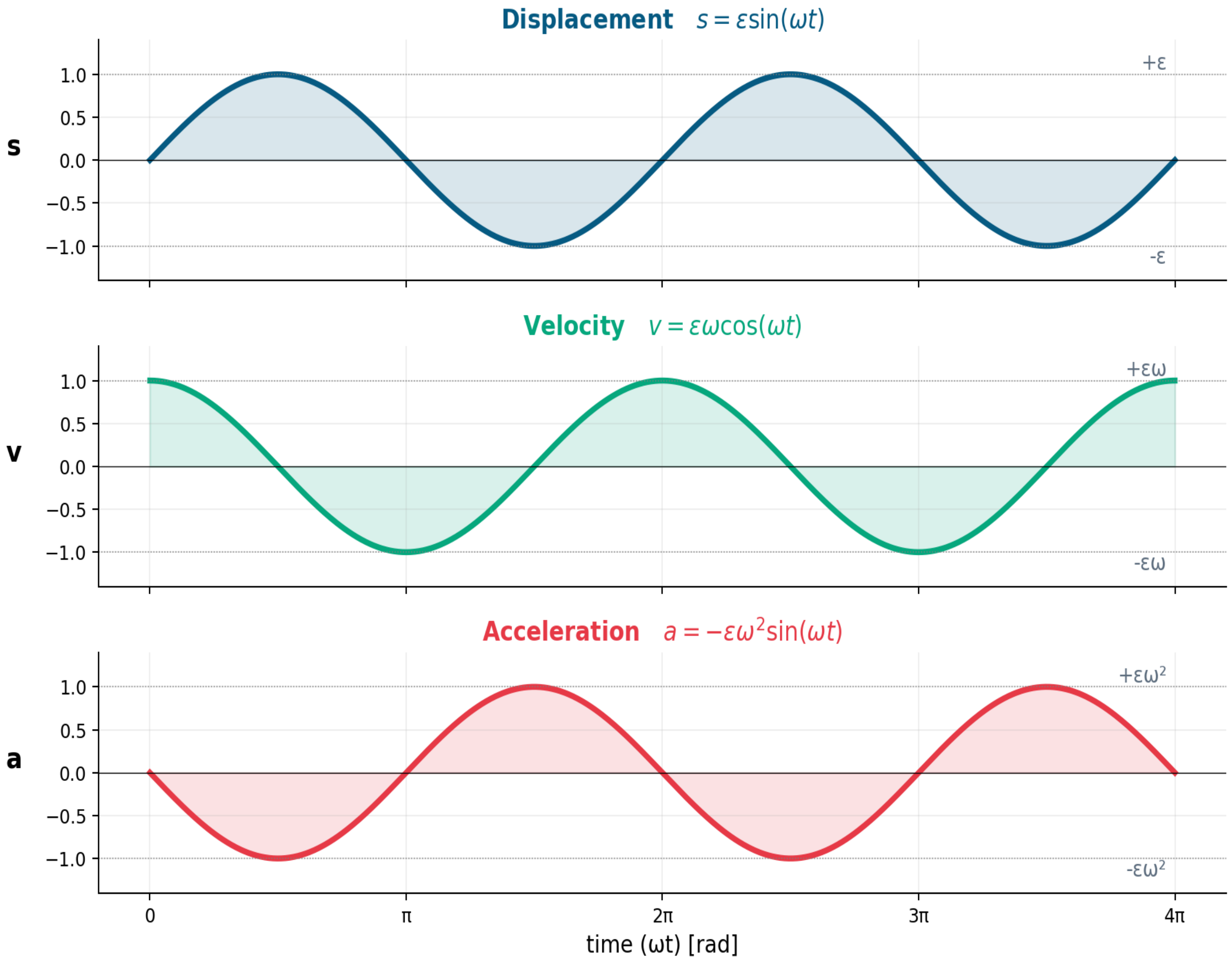
R	**Radius**	*Disc radius (cam body)*
ε	**Eccentricity**	*Offset of cam centre from shaft*
2ε	**Stroke**	*Total plate travel (peak-peak)*
ω	**Angular speed**	*Camshaft rotation (rad/s)*

When the shaft turns

the eccentric cam wobbles up and down, lifting the plate by ε in either direction. The plate executes pure Simple Harmonic Motion (SHM).

04

Mathematics of Plate Motion



Each plate's height vs time:

Displacement

$$s(t) = \varepsilon \sin(\omega t)$$

Unit: mm

Velocity

$$v(t) = \varepsilon\, \omega \cos(\omega t)$$

Unit: mm/s

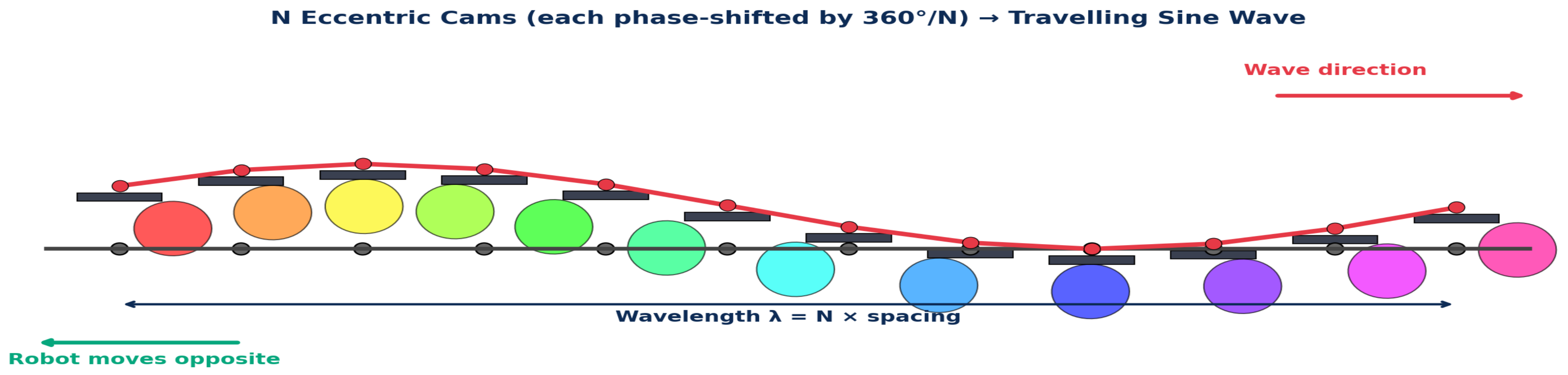
Acceleration

$$a(t) = -\varepsilon\, \omega^2 \sin(\omega t)$$

Unit: mm/s²

Peaks: $s_{max} = \varepsilon$ | $v_{max} = \varepsilon\omega$ | $a_{max} = \varepsilon\omega^2$

03 From One Cam to a Travelling Wave



1 **Stack N cams on one shaft**

Each cam is identical — just an eccentric circle. They are mounted along the shaft at equal spacing.

2 **Phase-shift each cam by 360°/N**

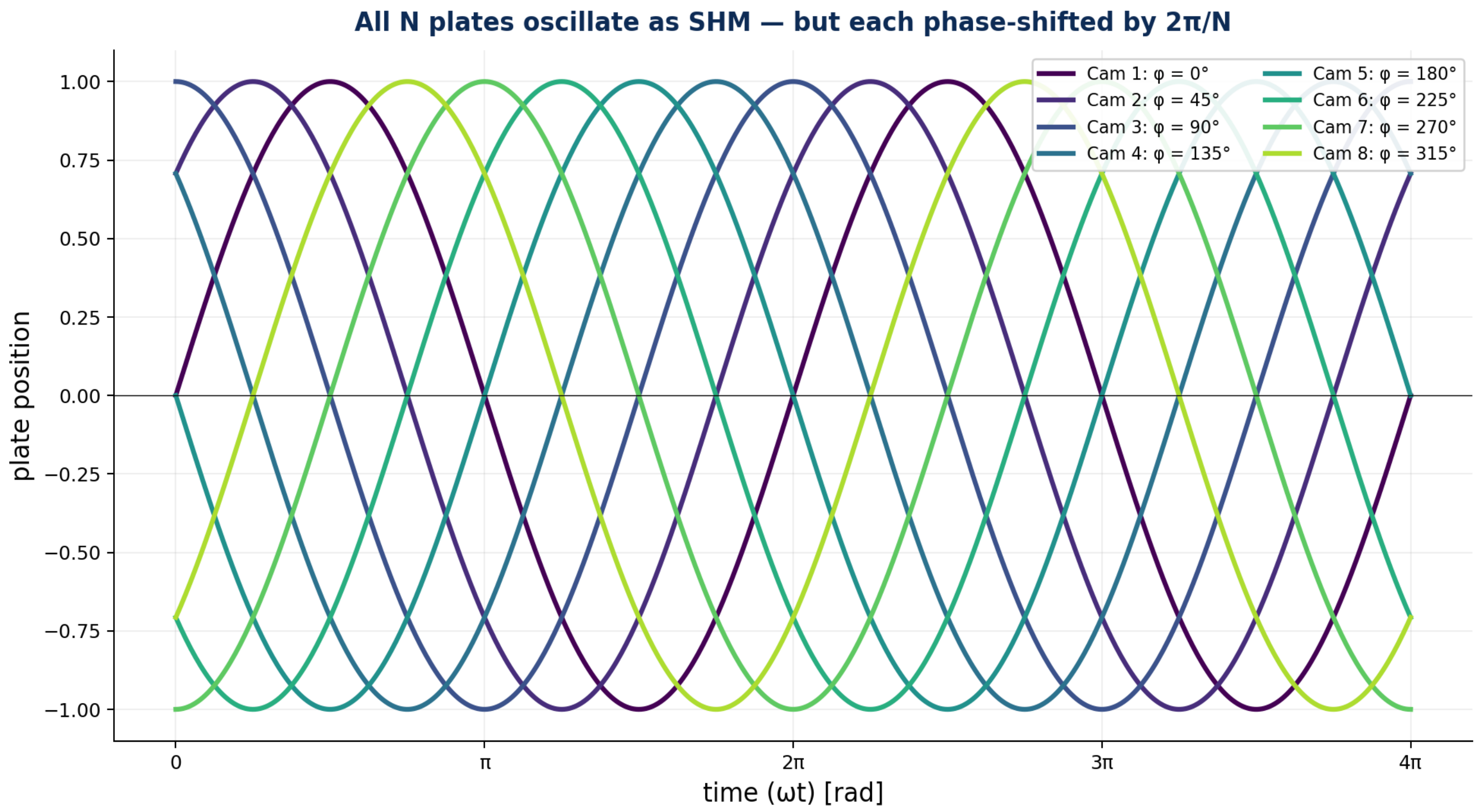
Cam #1 starts at 0°, cam #2 at 30°, cam #3 at 60° … cam #N completes a full cycle exactly along the shaft length.

3 **Result: a sine wave moving →**

As the shaft rotates, the wave travels along the chassis at speed v_wave = λ × frequency.

03

All Plates Are Sinusoidal — But Phase-Shifted



Phase Equation

$$y_i(t) = \varepsilon \sin(\omega t + (i-1) \cdot 2\pi/N)$$

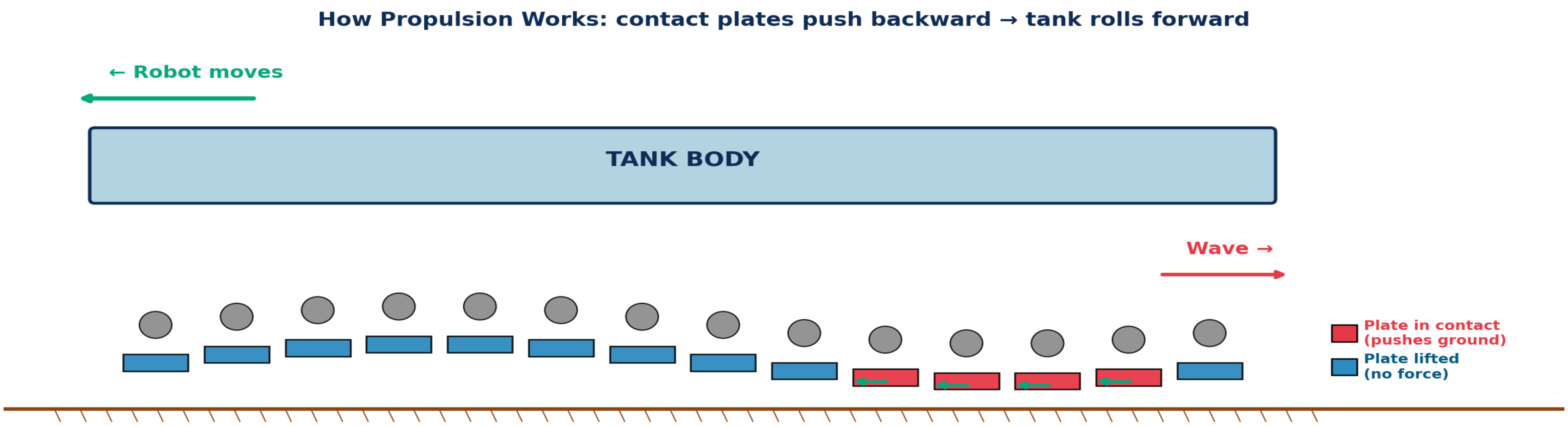
Where:

i = cam index (1, 2, …, N)
ε = eccentricity
ω = shaft angular velocity
N = number of cams

Each cam executes the same SHM, just delayed in time.

→ At any frozen instant, plotting the heights along the tank gives a sine wave.

Why Does the Tank Move?



Contact pattern

Only ~⅓ of plates touch the ground at any moment — those at the bottom of their cycle.

Wave moves backward

As the shaft rotates, the contact zone sweeps backward along the tank.

Tank moves forward

Newton's 3rd law: the ground pushes the tank forward via static friction.

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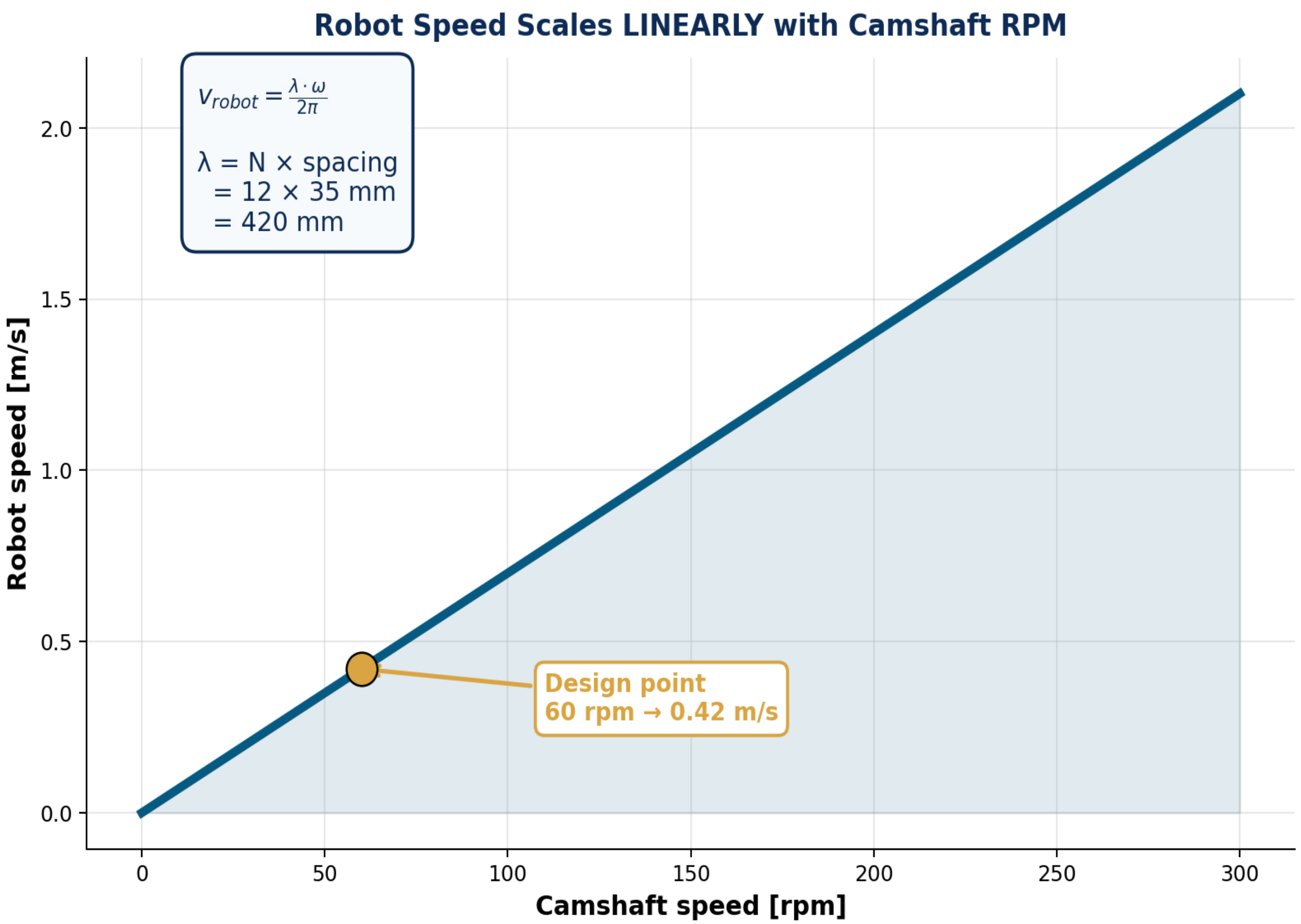
Speed & Performance Analysis



Design Trade-offs

More cams (N ↑)

Smoother wave, but heavier shaft & more parts

Larger ε

Bigger lift → more grip, but higher max acceleration

Faster RPM

Linear speed gain; limited by inertia & vibration

Cam radius R

Affects ground clearance; doesn't change wave shape

Linear speed control → ideal for variable-speed drive

MATLAB Simulation Suite

Five interactive simulations covering every aspect of the mechanism

1
Single Cam

ONE cam → SHM building block. Animated cam + scrolling s, v, a plots + phase plot.

2
Single vs Wave

Side-by-side: 1 cam (left) vs N cams (right) — see the wave form from individual cams.

3
2D Side View

Full mechanism: rotating shaft, all cams, all plates riding on top, robot body.

4
3D Tank View

Complete 3D model with terrain, tank body, cams, plates, and steering wheels.

5
Physics Analysis

Contact pattern, ghost-wave snapshots, raster of touching plates over time.

Run via: >> MAIN_wave_drive then choose [1]-[5] or [a] for all

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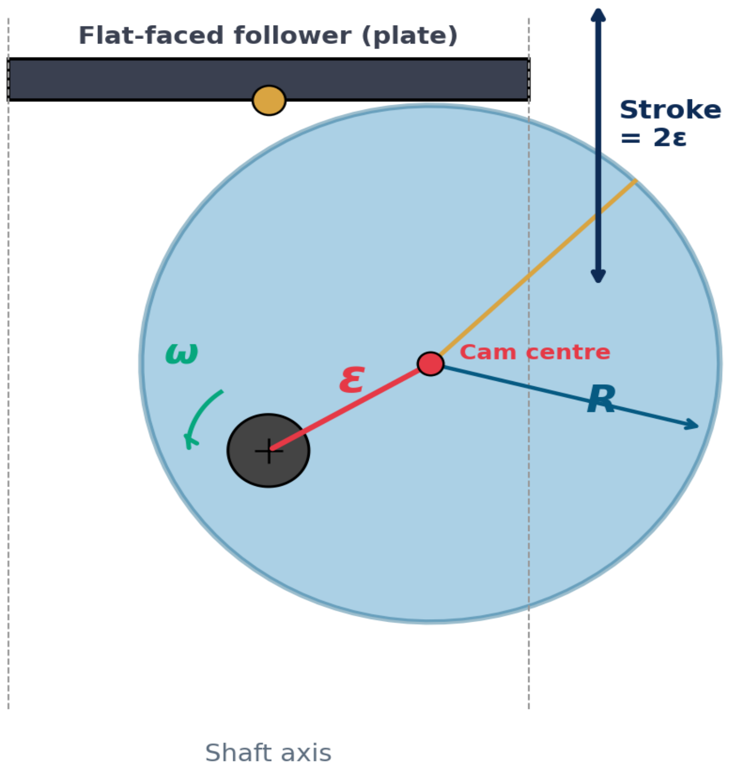
Reading the Single Cam Simulation



1 **Side view (large left panel)**

Shaft, eccentric cam, plate. Watch the cam wobble.

2 **Displacement s vs time**

Blue sinusoid that scrolls right-to-left like an oscilloscope.

3 **Velocity v vs time**

Green cosine — peaks where displacement is zero.

4 **Acceleration a vs time**

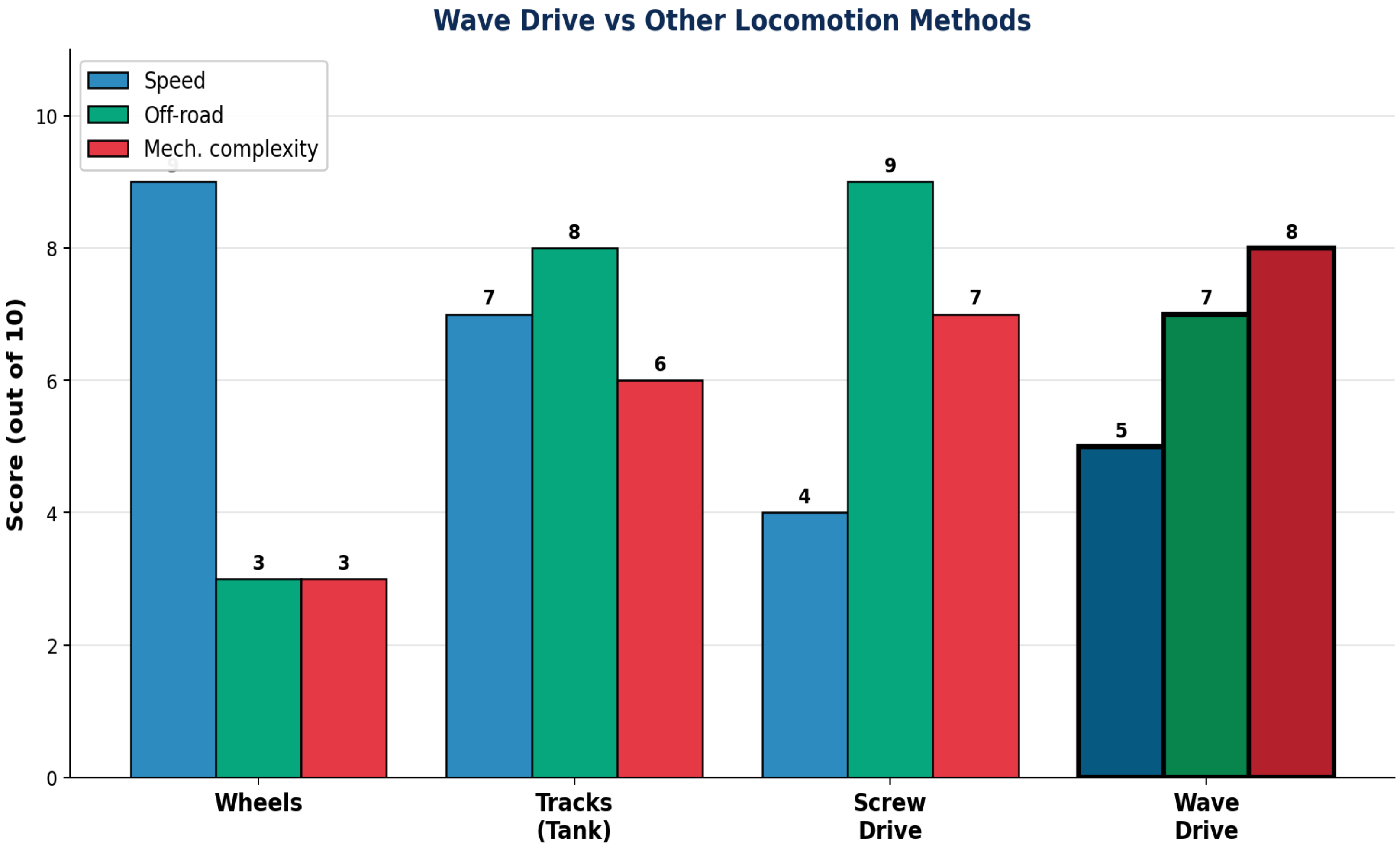
Red sinusoid (180° out of phase with s).

5 **Phase plot (v vs s)**

Purple ellipse — signature of pure SHM.

Tunable parameters (top of single_cam_simulation.m): cam_radius, eccentricity, n_rpm, total_time

07 Wave Drive vs Other Locomotion



Verdict

Best for

Soft / sandy / uneven terrain

Worst for

High-speed paved roads

Sweet spot

Slow off-road utility robots

Bonus

No exposed wheels — quieter and dust-resistant

08

Conclusions & Future Work

What we learned

✓ **Eccentric circular cam = simplest possible cam**

Just a circle off-centre on a shaft. SHM is automatic.

✓ **N phase-shifted cams create a travelling wave**

Mathematically equivalent to a sine wave with frequency = ω/(2π).

✓ **Robot speed scales linearly with shaft RPM**

v_robot = wavelength × frequency

✓ **Propulsion comes from contact friction**

Plates at the bottom of their cycle push the ground backward.

Future extensions

→ **Custom cam profiles**

Replace circular cams with cycloidal profiles → smoother lift, less vibration.

→ **Differential steering**

Two parallel camshafts driven independently → tank-like turning.

→ **Slip & friction modelling**

Real robot speed < wave speed when grip is limited.

→ Integration possibilities

Could be applied to off-road versions for sandy campuses & soft terrain.

Thank You

Questions & Discussion

Capital National University

Theory of Machines II | Sheet Project 2026